

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0514

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

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1. Grey Level Transformation

(a) Purpose

Grey level transformation is used to improve the visual appearance of an image by changing the intensity values of pixels. It helps in enhancing contrast, brightness, and image details.

(b) Negative Transformation

Formula:

$$s = (L - 1) - r$$

Where:

- s = Output pixel value
- r = Input pixel value
- $L = 256$ (8-bit image)

Steps

1. Take the original pixel value (r).
2. Substitute it into the formula.
3. Compute the new pixel value.

Example Table

Original Pixel (r)	Calculation	Output (s)
r	$255 - r$	s

2. Log Transformation

(a) Purpose

Log transformation expands dark pixel values and compresses brighter values. This makes details in dark regions more visible.

(b) Formula

$$s = c \log(1 + r)$$

Where:

- c = Constant
- r = Input pixel value

Steps

1. Add 1 to the pixel value.
2. Find the logarithm.
3. Multiply by constant c .

Calculation Table

Step	Operation
1	Calculate $(1 + r)$
2	Find $\log(1 + r)$
3	Multiply by c
4	Final Output = s

3. Histogram Processing

(a) Purpose

Histogram equalization improves image contrast by spreading the pixel intensity values over the available gray-level range.

(b) Given Histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Step 1: Total Number of Pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Step 2: Cumulative Distribution Function (CDF)

Intensity	Frequency	CDF
0	2	2
1	2	4
2	4	8
3	8	16

Step 3: Equalization Formula

$$\text{New Intensity} = (\text{CDF} / N) \times (L - 1)$$

For a 3-bit image,

$$L = 8$$

Final Equalized Values

Old Intensity	New Intensity
0	1
1	2
2	4
3	7

4. Spatial Filtering (Smoothing)

(a) Purpose

Spatial filtering removes noise by replacing each pixel with the average of its neighboring pixels.

(b) Averaging Filter

Filter	Value
All 9 values	1/9

Steps

1. Add all 9 neighboring pixel values.
 2. Divide the total by 9.
 3. The result is the new center pixel value.
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5. Spatial Filtering (Sharpening)

(a) Purpose

Sharpening enhances edges and fine image details.

(b) Laplacian Mask

0	-1	0
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-1	4	-1
0	-1	0

Steps

1. Multiply each mask value by the corresponding image pixel.
 2. Add all the values.
 3. The result is the sharpened center pixel.
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6. Frequency Domain Filtering

(a) Difference

Low-Pass Filter	High-Pass Filter
Keeps low frequencies	Keeps high frequencies
Removes noise	Enhances edges
Produces smooth image	Produces sharp image

(b)

Keep only the first two frequency components.

Example

Input:

F1, F2, F3, F4

Output:

F1, F2, 0, 0

7. Thresholding

(a) Purpose

Thresholding separates an image into foreground and background using a threshold value.

(b)

Rule

Condition	Output
$\text{Pixel} \geq \text{Threshold}$	1
$\text{Pixel} < \text{Threshold}$	0

Apply this rule to all pixel values.

8. Edge Detection

(a) Purpose

Edge detection identifies object boundaries where intensity changes rapidly.

(b) Horizontal Sobel Mask

-1 0 1

-2 0 2

-1 0 1

Steps

1. Multiply each mask value by the corresponding pixel.
 2. Add all the products.
 3. The sum gives the horizontal gradient.
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9. Region-Based Segmentation

(a) Purpose

Region growing groups neighboring pixels that have similar intensity values.

(b)

Seed Value = 50

Threshold Difference = 5

Rule

Condition	Result
$ \text{Pixel} - 50 \leq 5$	Include in Region
Otherwise	Exclude

10. Combined Enhancement and Segmentation

(a) Purpose

Smoothing removes noise before segmentation, improving segmentation accuracy.

(b)

Average Value = 48

Threshold Rule

Condition	Output
$\text{Pixel} \geq \text{Threshold}$	1
$\text{Pixel} < \text{Threshold}$	0

11. Power-Law (Gamma) Transformation

(a) Purpose

Gamma correction adjusts image brightness.

Gamma Value	Effect
$\gamma < 1$	Brightens image
$\gamma = 1$	No change
$\gamma > 1$	Darkens image

(b) Formula

$$s = c \times r^{\gamma}$$

Steps

1. Take the input pixel value.
 2. Raise it to the power γ .
 3. Multiply by c .
 4. Obtain the output value.
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12. Contrast Stretching

(a) Purpose

Contrast stretching improves image contrast by expanding the intensity range.

(b) Formula

$$s = ((r - r_{\min}) / (r_{\max} - r_{\min})) \times 255$$

Where:

- $r_{\min} = 50$
- $r_{\max} = 150$

Steps

1. Subtract 50.
2. Divide by 100.

3. Multiply by 255.
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13. Histogram Equalization (4-bit Image)

(a) Purpose

Histogram equalization spreads pixel intensities across the available gray-level range to improve image contrast.

(b) Given Histogram

Intensity	Frequency
0	4
1	6
2	10
3	6

Cumulative Distribution

Intensity	Frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

Equalized Intensities

Old Intensity	New Intensity
0	2
1	6

2	12
3	15

14. Median Filtering

(a) Purpose

Median filtering removes salt-and-pepper noise while preserving edges.

Steps

1. Arrange all 9 pixel values in ascending order.
 2. Select the middle value (median).
 3. Replace the center pixel with the median.
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15. Gaussian Smoothing

(a) Purpose

Gaussian smoothing reduces image noise while preserving edges better than averaging.

(b) Gaussian Kernel

1 2 1

2 4 2

1 2 1

Kernel Sum = 16

Steps

1. Multiply each kernel coefficient by the corresponding pixel value.
2. Add all the products.
3. Divide the total by 16.
4. The result is the new center pixel.